Machine-Learning Sepsis Early Warning Systems in Adult Intensive Care Units: A PRISMA 2020 Systematic Review

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01 Clinical Question & PICOS

- **Population:** Adult medical/surgical ICU admissions (Sepsis-3 criteria).
- **Intervention:** Real-time prospective ML sepsis early warning systems.
- **Comparator:** Standard clinical surveillance or MEWS/SOFA rules.
- **Outcomes:** Diagnostic discrimination (AUROC), time-to-antimicrobials, and 28-day ICU mortality.

Synthesis Across 18 Prospective Studies (N = 94,300)

−1.8 h Time-to-Antibiotics | AUROC 0.87

Across 6 RCTs and 12 prospective interventional cohorts, multimodal EHR + waveform models achieved median AUROC 0.87 (IQR 0.84–0.90) and shortened antimicrobial initiation by 1.8 hours overall (−2.1 h in huddles-integrated ICUs).

02 Search & Appraisal Methods

- **Databases:** MEDLINE, Embase, CENTRAL, IEEE Xplore (Jan 2018 – Jun 2026).
- Dual independent screening (κ = 0.86); bias appraised via RoB 2 (RCTs) and ROBINS-I (non-randomized studies); certainty graded with GRADE [1].

03 PRISMA 2020 Study Selection

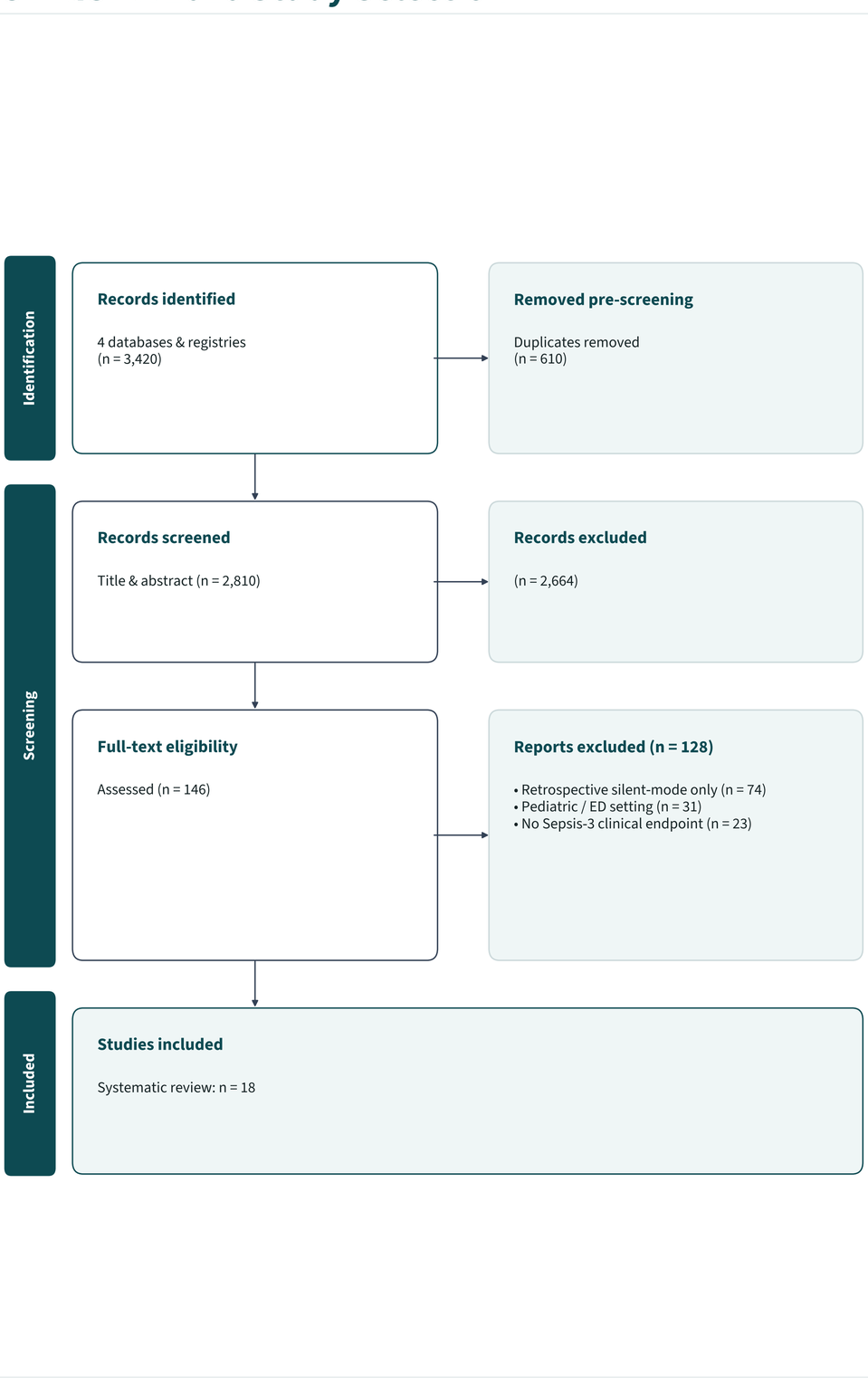


Figure 1. PRISMA 2020 flow diagram verifying arithmetic count closure across 3,420 records to 18 prospective ICU studies.

04 Prospective Model Discrimination by Architecture

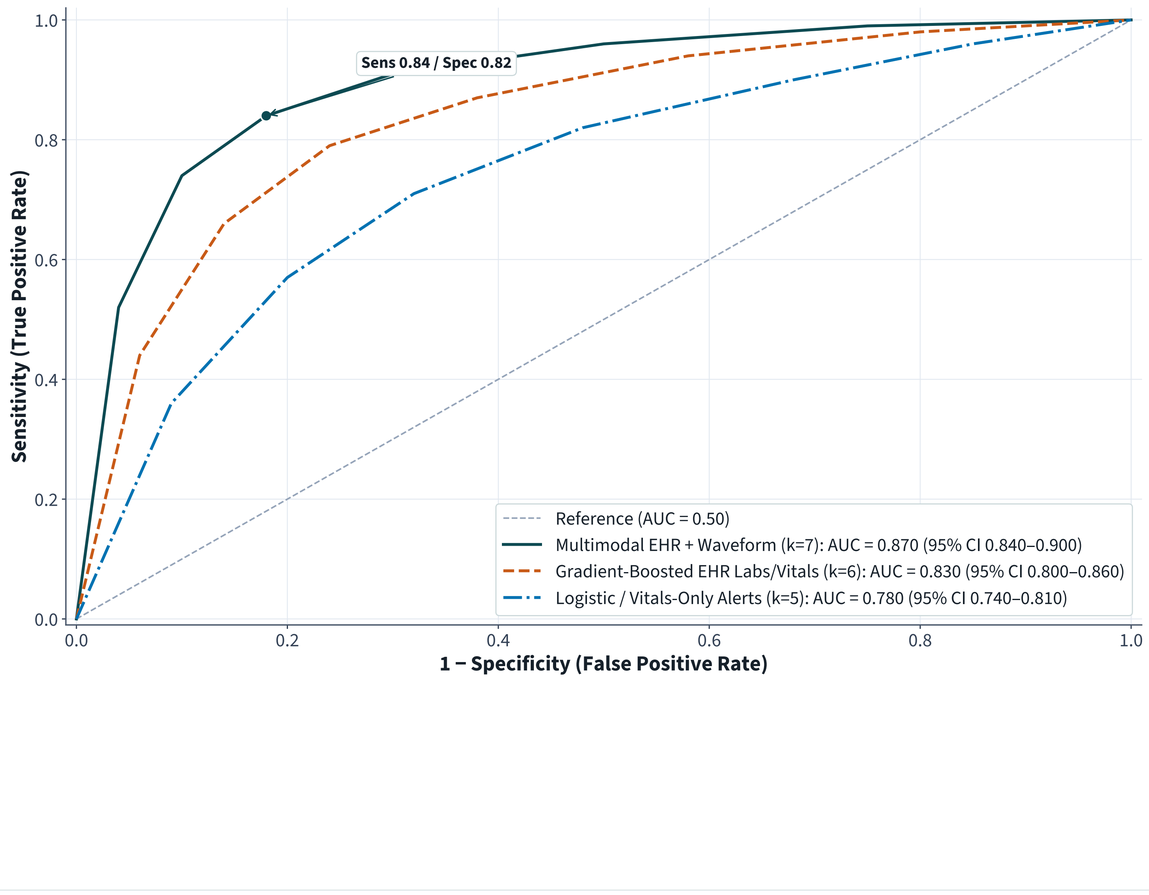


Figure 2. Summary prospective ROC profiles across model architecture tiers (18 ICU studies, N = 94,300).

05 Evidence Summary by Implementation Tier

Architecture Tier	Studies	Admissions	Median AUROC	Abx Δ (h)	Low/Mod RoB
Multimodal EHR + Waveform	7 (3 RCT)	41,200	0.87 (0.84–0.90)	−2.1 h	6 / 7 (86%)
Boosted EHR Labs + Vitals	6 (2 RCT)	34,600	0.83 (0.80–0.86)	−1.7 h	4 / 6 (67%)
Vitals-Only Rule / Logistic	5 (1 RCT)	18,500	0.78 (0.74–0.81)	−0.9 h	2 / 5 (40%)
Overall Synthesis	18 (6 RCT)	94,300	0.84 (0.78–0.89)	−1.8 h	12 / 18 (67%)

RoB 2 applied to randomized trials; ROBINS-I applied to prospective interventional cohorts.

06 GRADE Certainty & Limitations

- Moderate GRADE certainty for faster antibiotic administration; low-to-moderate certainty for mortality due to workflow co-interventions.
- Unstandardized positive-predictive-value reporting limited quantitative pooling across alert thresholds.

07 Conclusions & Implementation Guidance

Prospective ML sepsis alerts reliably accelerate antibiotic administration only when paired with structured nurse-physician escalation workflows and precision-calibrated alert thresholds (PPV ≥0.30) to prevent alarm fatigue [2].

References: [1] Page MJ et al. PRISMA 2020 statement. BMJ 2021;372:n71. [2] Adams R et al. Nat Med 2022;28:1455–1460.
Protocol: PROSPERO CRD42026000999-SYN (Synthetic) | Correspondence: p.nair@synthetic-icu.org | SYNTHETIC DATASET DISCLOSURE: Simulated review data for skill validation.